

English	(1015CMD303029230004)		Test Pattern
		CLASSROOM CONTACT PROGRAMME (Academic Session : 2023-2024)	NEET (UG)
			MINOR
			21-05-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MEB-I (A)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill 0012345678.

Test Booklet Code

This Booklet contains 28 pages.

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Do not open this Test Booklet until you are asked to do so.

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Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
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- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
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- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

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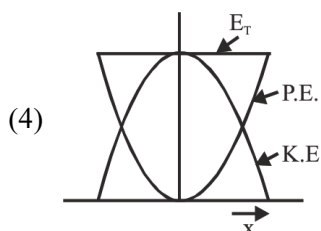
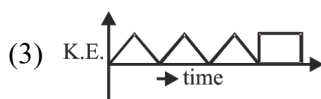
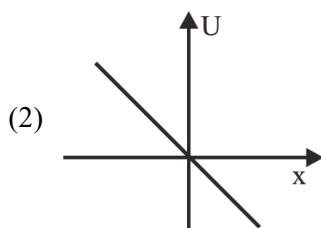
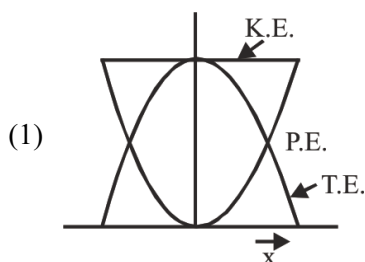
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SECTION-A (PHYSICS)

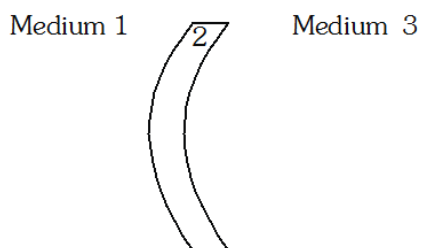
1. A beam of light of wavelength 600nm from a distant source falls on a single slit 1mm wide and the resulting diffraction pattern is observed on a screen 2m away. The distance between first dark fringe on either side of central bright fringe is -
(1) 1.2cm (2) 1.2mm (3) 2.4cm (4) 2.4mm
2. Wavelength of light used in an optical instrument are $\lambda_1 = 4000 \text{ \AA}$ and $\lambda_2 = 5000 \text{ \AA}$, then ratio of their respective resolving powers (corresponding to λ_1 and λ_2) is-
(1) 16 : 25 (2) 9 : 1 (3) 4 : 5 (4) 5 : 4
3. An unpolarised beam of intensity I_0 is incident on a pair of nicols making an angle of 60° with each other. The intensity of light emerging from the pair is :-
(1) I_0 (2) $I_0/2$ (3) $I_0/4$ (4) $I_0/8$
4. Which is correct graph for the particle performing SHM :-



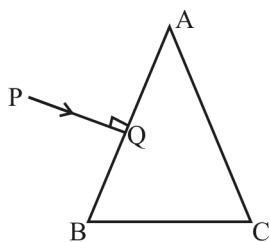
5. When the angle of incidence on a material from air is 60° , the reflected light is completely polarized. The velocity of the refracted ray inside the material is (in ms^{-1}) :-
(1) 3×10^8
(2) $\left(\frac{3}{\sqrt{2}}\right) \times 10^8$
(3) $\sqrt{3} \times 10^8$
(4) 0.5×10^8
6. A particle executes simple harmonic motion between $x = -A$ and $x = +A$. At $x = 0$ it is moving in $+x$ -direction. The time taken for it to move from $x = 0$ to $x = \frac{A}{2}$ is T_1 and to move from $\frac{A}{2}$ to $\frac{A}{\sqrt{2}}$ is T_2 , then
(1) $T_1 < T_2$ (2) $T_1 = T_2$
(3) $T_2 = 2T_1$ (4) $T_1 = 2T_2$
7. The instantaneous displacement of a simple pendulum oscillator is given by $x = A \cos\left(\omega t + \frac{\pi}{4}\right)$. Its speed will be maximum at time :-
(1) $\frac{\pi}{4\omega}$
(2) $\frac{\pi}{2\omega}$
(3) $\frac{\pi}{\omega}$
(4) $\frac{2\pi}{\omega}$
8. A concave mirror of focal length f produces an image n times the size of the object. If the image is real, then the distance of the object from the mirror is :
(1) $(n-1)f$
(2) $[(n-1)/n]f$
(3) $[(n+1)/n]f$
(4) $(n+1)f$

9. The focal length of the lens is nR as shown in the figure. If the radii of curvature of both the surfaces are equal to R , then value of n is :

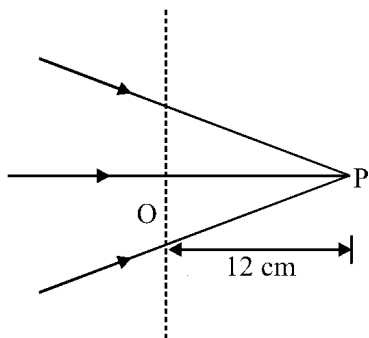
($\mu_1 = 1$, $\mu_2 = 0.5$ and $\mu_3 = 2$)



- (1) 1 (2) 2
(3) 1.5 (4) 2.5
10. As shown in figure, PQ is the ray incident on a prism ABC. The critical angle for the material of the prism is 45° . Find the refractive index of the material of the prism. :-



- (1) 1.5 (2) 2
(3) $\sqrt{2}$ (4) 1.33
11. Figure given below shows a beam of light converging at point P. When a convex lens of focal length 16 cm is introduced in the path of the beam at a place O shown by dotted line such that OP becomes the axis of the lens, the beam converges at a distance x from the lens. The value x will be equal to :-



- (1) 12 cm (2) 24 cm
(3) 48 cm (4) 6.8 cm

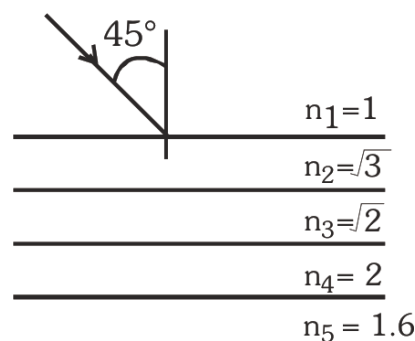
12. In a compound microscope the objective lens and eye-piece have focal length of 0.95 cm and 5 cm respectively, and are kept at a distance of 20 cm. The last image is formed at a distance of 25 cm from the eye-piece calculate magnifying power:-

- (1) 94 (2) 84
(3) 75 (4) 88

13. Keeping incident ray fixed a plane mirror is rotated with an angle of 20° . Then from what angle reflected ray must rotate.

- (1) 10° (2) 20°
(3) 40° (4) 60°

14. In the figure shown the angle made by the light ray with the normal in the medium of refractive index $\sqrt{2}$ is :-



- (1) 30° (2) 60°
(3) 90° (4) None of these

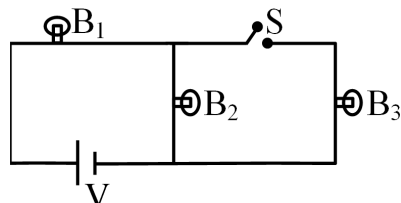
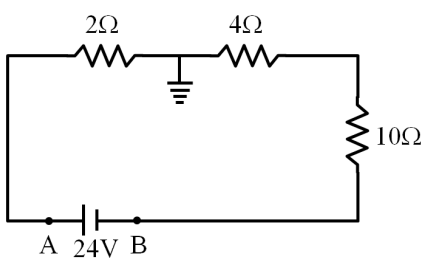
15. A fish at a depth of 12 cm in water is viewed by an observer. To what depth will its image appear ?

$$\left(\mu_w = \frac{4}{3}\right)$$

- (1) 3 cm
(2) 12 cm
(3) 6 cm
(4) 9 cm

16. A ray of light propagates from glass ($\mu_g = 3/2$) to water ($\mu_w = 4/3$). The value of the critical angle.

- (1) $\sin^{-1}(1/2)$ (2) $\sin^{-1}(\sqrt{8}/9)$
(3) $\sin^{-1}(8/9)$ (4) $\sin^{-1}(5/7)$

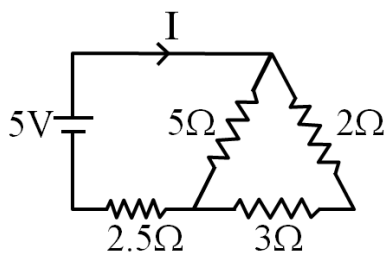
17. An object placed at a distance of 9 cm from first principal focus of convex lens, produces a real image at a distance of 25 cm from its second principal focus. Then focal length of lens is
- (1) 9 cm (2) 25 cm
(3) 15 cm (4) 17 cm
18. Two SHMs are given by $y_1 = a \sin \left[\left(\frac{\pi}{2} \right) t + \phi \right]$ and $y_2 = b \sin \left[\left(\frac{2\pi t}{3} \right) + \phi \right]$. The phase difference between them after 1 sec. is :-
- (1) π (2) $\pi/2$
(3) $\pi/4$ (4) $\pi/6$
19. Starting from the origin a body oscillates simple harmonically with a period of 2 second. After what time will its kinetic energy be 75% of the total energy?
- (1) $\frac{1}{3}$ sec. (2) $\frac{1}{12}$ sec.
(3) $\frac{1}{6}$ sec. (4) $\frac{1}{4}$ sec.
20. The maximum velocity of a particle executing simple harmonic motion with an amplitude 7 mm, is 4.4 m/s. The period of oscillation is :-
- (1) 0.1 sec (2) 100 sec
(3) 0.01 sec (4) 10 sec
21. The average acceleration in one time period in a simple harmonic motion is :-
- (1) $A\omega^2$ (2) $\frac{A\omega^2}{2}$
(3) $\frac{A\omega^2}{\sqrt{2}}$ (4) Zero
22. A particle executes SHM according to equation $x = 10(\text{cm}) \cos \left[2\pi t + \frac{\pi}{2} \right]$, where t is in seconds. The magnitude of the velocity of the particle at $t = \frac{1}{6}$ sec. will be :-
- (1) 24.7 cm/s (2) 20.5 cm/s
(3) 28.3 cm/s (4) 31.4 cm/s
23. A car battery of emf 12 V and internal resistance $5 \times 10^{-2} \Omega$, receives a current of 60 A, from external source, then terminal potential difference of battery is
- (1) 12 V (2) 9 V
(3) 15 V (4) 20 V
24. A (100 W, 200 V) bulb is connected to a 160 volt supply. The power consumption would be
- (1) 100 W
(2) 125 W
(3) 64 W
(4) 80 W
25. Three identical bulbs B_1 , B_2 and B_3 are connected to the mains as shown in figure. If B_3 is disconnected from the circuit by opening switch S, then incandescence of bulb B_1 will
- 
- (1) Increase
(2) Decrease
(3) Become zero
(4) No change
26. In the given circuit potential of point A and B are respectively-
- 
- (1) +24V, zero
(2) +24V, -24V
(3) +3V, -21V
(4) -3V, +21V

English

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27. The current I through the cell in the network shown is



- (1) 3 A (2) 2 A (3) 1 A (4) 4 A

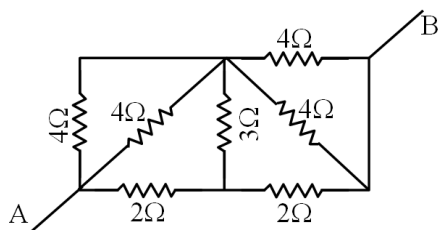
28. **Assertion :** The voltage across a battery may be less, equal or more than the emf of the battery.

Reason : Voltage across a battery also depends on the magnitude and direction of current

- (1) Both Assertion & Reason correct but Reason is not the correct explanation of Assertion.
 (2) Both Assertion & Reason correct and Reason is the correct explanation of Assertion.
 (3) Assertion is not correct but Reason is correct.
 (4) Assertion is correct but Reason is not correct.

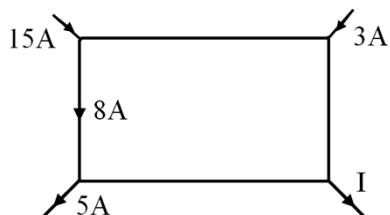
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29. Resistance across AB as shown in figure is



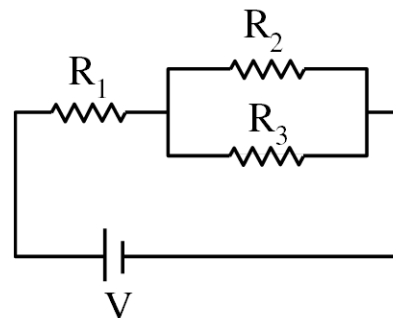
- (1) 2 Ω (2) 4 Ω
 (3) 6 Ω (4) 12 Ω

30. The value of current I in the given circuit is –



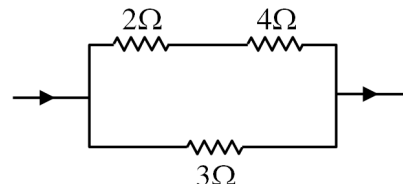
- (1) 3A (2) 13A
 (3) 23A (4) -3A

31. Three identical resistors $R_1 = R_2 = R_3$ are connected as shown to a battery of constant e.m.f. The power dissipated is



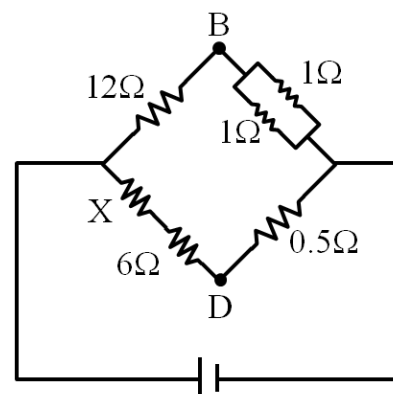
- (1) The least in R_1
 (2) Greatest in R_1
 (3) In the ratio 1 : 2 in resistance R_1 and R_2 respectively
 (4) The same in R_1 and in the parallel combination of R_2 and R_3

32. In a circuit shown in figure, the power delivered in 3 ohm resistor due to a current flowing in it is 12 J. The power delivered in 4 ohm resistor is :



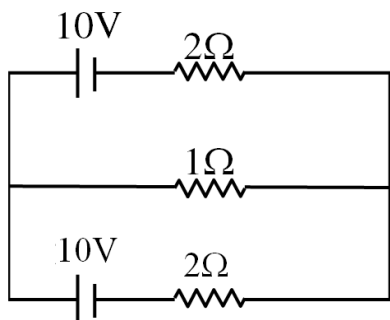
- (1) 2 W (2) 4 W (3) 64 W (4) 32 W

33. For what value of unknown resistance X the potential difference between B and D will be zero in arrangement of adjacent figure–



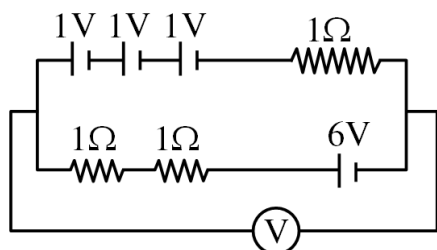
- (1) 4Ω (2) 3Ω
 (3) 6Ω (4) 2Ω

34. The value of current through $2\ \Omega$ resistor is



- (1) 10 A (2) 2.5 A
(3) 5 A (4) 2 A

35. Reading of an ideal voltmeter in the circuit below is



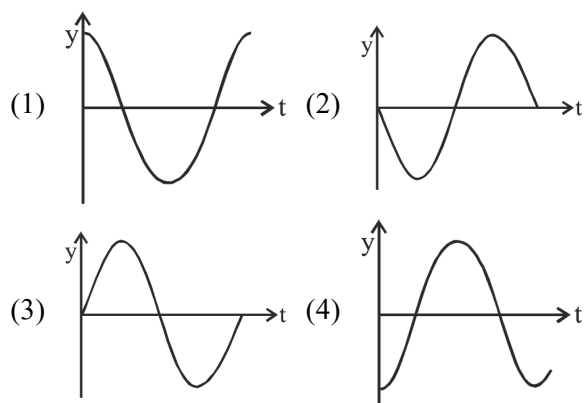
- (1) Zero (2) 2 V
(3) 4 V (4) 6 V

SECTION-B (PHYSICS)

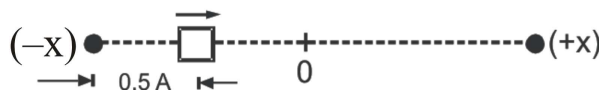
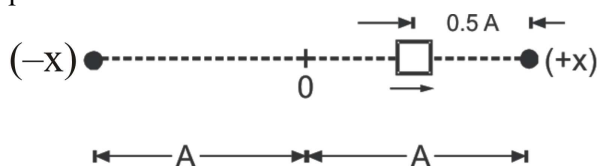
36. A particle executes SHM with a time period of 4 sec and amplitude 5 cm. Maximum magnitude of its velocity is :-

- (1) 5π cm/s (2) 10π cm/s
(3) 2.5π cm/s (4) 5.5π cm/s

37. If in a linear SHM, time is counted from positive extreme position, then which of the given graphs will demonstrate this SHM?



38. Two bodies performing S.H.M. have same amplitude and frequency. Their phases at a certain instant are as shown in the figure. The phase difference between them is



- (1) $\frac{11}{6}\pi$ (2) π
(3) $\frac{\pi}{3}$ (4) $\frac{3}{5}\pi$

39. A person is suffering from myopic defect. He is able to see clear objects placed at 15 cm. What type and of what focal length of lens he should use to see clearly the object placed 60 cm away

- (1) Concave lens of 20 cm focal length
(2) Convex lens of 20 cm focal length
(3) Concave lens of 12 cm focal length
(4) Convex lens of 12 cm focal length

40. Focal length of objective and eye piece in astronomical telescope is 50 cm and 2 cm. For normal adjustment magnifying power and length of the telescope will be :-

- (1) 25, 50 cm
(2) 25, 52 cm
(3) 25, 54 cm
(4) 25, 56 cm

41. A ray of light incidents on a plane mirror at an angle of 30° . The deviation produced in the ray is :-

- (1) 30°
(2) 60°
(3) 90°
(4) 120°

42. A prism is made up of a material of refractive index $\sqrt{3}$. The angle of prism is A . If the angle of minimum deviation is equal to the angle of the prism then the value of A is :-

(1) 30°
 (2) 60°
 (3) 45°
 (4) 75°

43. Two lenses having $f_1 : f_2 = 2 : 3$ has combination to make no chromatic aberration. Find the ratio of dispersive power of glasses used :-

(1) $2 : 3$
 (2) $3 : 2$
 (3) $4 : 9$
 (4) $9 : 4$

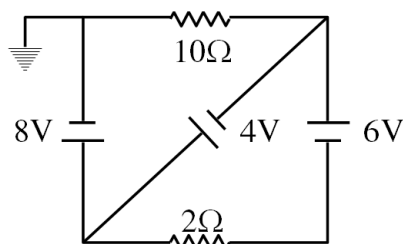
44. A simple harmonic motion has an amplitude A and time period T . The time required by it to travel from $x = A$ to $x = A/2$ is :-

(1) $T/6$ (2) $T/4$
 (3) $T/3$ (4) $T/2$

45. A particle moves in a circular path with a continuously increasing speed. Its motion is :-

(1) Periodic
 (2) Oscillatory
 (3) Simple harmonic
 (4) None of these

46. In the circuit shown in figure, all cells are ideal. The current through $2\ \Omega$ resistor is

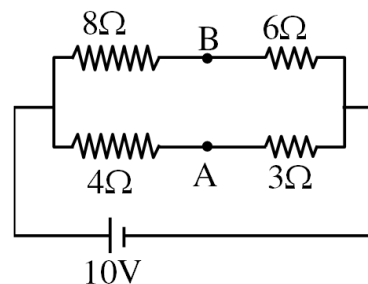


(1) 5 A (2) 1 A
 (3) 0.2 A (4) Zero

47. A cell sends a current through a resistance R for time t . Now the same cell sends current through another resistance r for the same time. If same amount of heat is developed in both the resistance, then the internal resistance of cell is

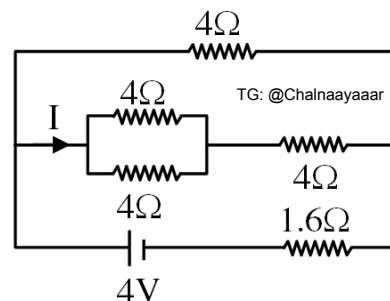
(1) $\frac{(R+r)}{2}$ (2) $\frac{(R-r)}{2}$
 (3) $\frac{(Rr)}{2}$ (4) $\sqrt{(Rr)}$

48. The potential difference between points A and B is



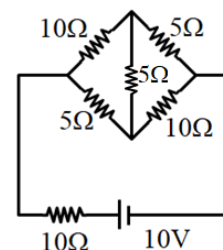
(1) $\frac{20}{7}$ V (2) $\frac{40}{7}$ V
 (3) $\frac{10}{7}$ V (4) Zero

49. In the circuit shown the value of I in ampere is



(1) 1 (2) 0.6
 (3) 0.4 (4) 1.5

50. Find current flow from battery-



(1) $\frac{4}{17}$ A (2) $\frac{10}{17}$ A
 (3) $\frac{6}{17}$ A (4) $\frac{5}{17}$ A

Topic : COORDINATION COMPOUNDS, p-BLOCK (TILL TAUGHT), COMMERCIAL BATTERIES, SOLID STATE, SURFACE CHEMISTRY [BACK UNIT] : CHEMICAL KINETICS

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SECTION-A (CHEMISTRY)

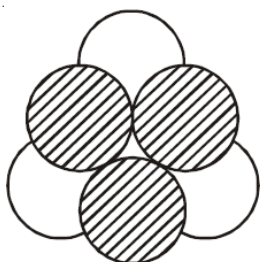
51. Which of the following crystal is represented by $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90^\circ$?

- (1) Orthorhombic (2) Monoclinic
(3) Triclinic (4) Tetragonal

52. The radius of the Na^+ is 95 pm and that of Cl^- ion is 181 pm. The co-ordination number of Na^+ will be

- (1) 4 (2) 6
(3) 8 (4) Unpredictable

53. The empty space between the shaded balls and hollow balls as shown in the diagram is called



- (1) Hexagonal void
(2) Octahedral void
(3) Tetrahedral void
(4) Double triangular void

54. Which of the following is incorrect ?

- (1) Tyndall effect provides information about size and shape of disperse particles.
(2) Mechanical dispersion is used to prepare printer ink.
(3) Coagulating power of an ion depends on both size and charge of ion.
(4) Both smog and fog are aerosol.

55. Metal excess defect due to anionic vacancies is observed in:

- (1) AgBr (2) NaCl
(3) ZnO (4) Metallic solids

56. In solid lattice, the cation has left a site and is located at interstitial position, the lattice defect is called

- (1) Interstitial defect (2) Vacancy defect
(3) Frenkel defect (4) Schottky defect

57. The number of atoms present in a unit cell of a monoatomic element of a simple cubic lattice, body-centred cubic and face centred cubic respectively are

- (1) 8, 9 and 14 (2) 1, 2 and 4
(3) 4, 5 and 6 (4) 2, 3 and 5

58. When the substance is placed in magnetic field all the domains get oriented in the direction of magnetic field. Substance is:

- (1) Paramagnetic (2) Ferromagnetic
(3) Ferrimagnetic (4) Antiferromagnetic

59. Experimentally it was found that a metal oxide has formula $\text{M}_{0.98}\text{O}_1$. Metal M, is present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be :-

- (1) 7.01% (2) 4.08% (3) 6.05% (4) 8.12%

60. In the face centred cubic lattice, atom A occupies the corner positions and atom B occupies the face centre positions. If one atom of B is missing from one of the face centred points, the formula of the compound is

- (1) A_2B (2) AB_2 (3) A_2B_2 (4) A_2B_5

61. On adding AgNO_3 solution into KI solution, a negatively charged colloidal sol is obtained when they are in :-

- (1) 50 mL of 0.1 M AgNO_3 + 50 mL of 0.1 M KI
(2) 50 mL of 0.1 M AgNO_3 + 50 mL of 0.2 M KI
(3) 50 mL of 0.2 M AgNO_3 + 50 mL of 0.1 M KI
(4) None of these

62. Which of the following acts as an emulsifier?

- (1) Soap (2) Water
(3) CH_3COOH (4) NaCl

63. How many of the following statements are correct:

- (a) Physical adsorption can be converted into chemical adsorption at high temperature.
(b) Heterogenous catalysis is an example of chemisorption.
(c) A catalyst can catalyse a non spontaneous reaction.
(d) Chemisorption is irreversible.

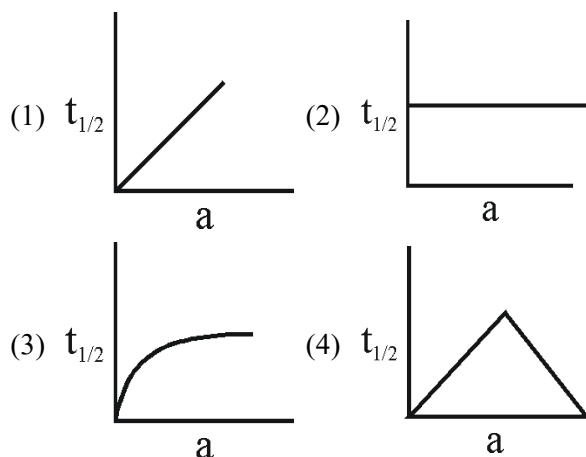
- (1) 1 (2) 2 (3) 3 (4) 4

64. For 1st order reaction find ratio of $t_{99.9\%}$ to that of $t_{50\%}$.

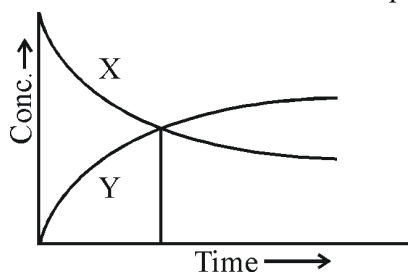
- (1) 10 (2) 3 (3) 4 (4) 17

65. Which of the following curves represents a 1st order reaction-

($t_{1/2}$ = half life), (a = initial concentration)



66. The accompanying figure depicts the change in concentration of species X and Y for the reaction $\text{X} \rightarrow \text{Y}$, as a function of time. The point of intersection of the two curves represents :



- (1) $t_{1/2}$
(2) $t_{3/4}$
(3) $t_{2/3}$
(4) Data is insufficient to predict

67. An endothermic reaction $\text{A} \rightarrow \text{B}$ have an activation energy of 15 kcal/mol and the heat of the reaction is 5 kcal/mol. The activation energy of the reaction $\text{B} \rightarrow \text{A}$ is :

- (1) 20 kcal/mol (2) 15 kcal/mol
(3) 10 kcal/mol (4) zero

68. The rate of the elementary reaction, $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$, when the volume of the reaction vessel is doubled :

- (1) will grow eight times of its initial rate
(2) reduce to one-eighth of its initial rate
(3) will grow four times of its initial rate
(4) reduce to one-fourth of its initial rate

69. The concentration of a reactant in a solution falls from 0.2 M to 0.1 M in 2 hours and from 0.2 M to 0.05 M in 4 hours. The order of the reaction is—

- (1) Zero (2) Two (3) One (4) Half

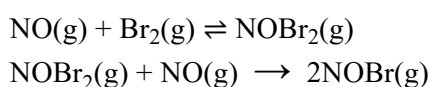
70. Which of the following is not a characteristics of lyophilic colloids :

- (1) Viscosity is higher than dispersion medium
(2) Colloid particles are heavily hydrated
(3) Irreversible in nature
(4) They are easily formed by direct mixing

71. The rate constant for the first order reaction is given by : $\log K = 14.34 - \frac{1.25 \times 10^4}{T}$ Calculate the activation energy :

- (1) $2.393 \times 10^5 \text{ J}$ (2) $8.69 \times 10^4 \text{ J}$
(3) $1.18 \times 10^5 \text{ J}$ (4) $8.69 \times 10^5 \text{ J}$

72. The following mechanism has been proposed for the reaction of NO with Br_2 to form NOBr :



If the second step is the rate determining step, the order of the reaction with respect to NO(g) is

- (1) 2 (2) 1 (3) 0 (4) 3

73. **Assertion (A)** : The formation of micelles takes place only above a particular concentration called CMC

Reason (R) : On dilution micelles revert back to ions.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

74. Which set of d-orbitals is involved in hybridisation of complex $[\text{Cr}(\text{NH}_3)_6]^{+3}$

- (1) $d_{xz}, d_{x^2-y^2}$ (2) d_z^2, d_{xz}
- (3) $d_z^2, d_{x^2-y^2}$ (4) d_{xy}, d_{yz}

75. For high spin octahedral complex Jahn-Teller effect is not observed in :-

- (1) d^4 (2) d^7 (3) d^9 (4) d^8

76. The IUPAC name of the Wilkinson's catalyst $[\text{RhCl}(\text{PPh}_3)_3]$ is :

- (1) Chlorotris(triphenylphosphine)rhodium (I)
- (2) Chlorotris(triphenylphosphine)rhodium (IV)
- (3) Chlorotris(triphenylphosphine)rhodium (0)
- (4) Chlorotris(triphenylphosphine)rhodium (VI)

77. The shape of complex $[\text{Ag}(\text{NH}_3)_2]^+$ is :

- (1) Octahedral (2) Square planar
- (3) Tetrahedral (4) Linear

78. Which type of ligand is contained in linkage isomerism?

- (1) Monodentate (2) Bidentate
- (3) Polydentate (4) Ambidentate

79. Match the correct option :

Column-I		Column-II	
(a)	Zeigler-natta catalyst	(i)	$[\text{RhCl}(\text{PPh}_3)_3]$ TG: @Chalnaayaaar
(b)	Cis-platin	(ii)	$\text{Al}(\text{C}_2\text{H}_5)_3 + \text{TiCl}_4$
(c)	Wilkinson catalyst	(iii)	B_{12}
(d)	Cobalt	(iv)	$\text{Cis}[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

- (1) a-ii, b-iv, c-i, d-iii
- (2) a-ii, b-i, c-iv, d-iii
- (3) a-iii, b-i, c-ii, d-iv
- (4) a-iv, b-iii, c-ii, d-i

80. For lead poisoning the antidote used is

- (1) $[\text{Ag}(\text{CN})_2]^-$
- (2) Cis -Platin
- (3) Nickel
- (4) $[\text{Ca}(\text{EDTA})]^{2-}$

81. If $\Delta_0 < P$ then correct configuration for d^4 would be –

- (1) $t_{2g}^3 e_g^1$ (2) $t_{2g}^4 e_g^0$
- (3) $t_{2g}^2 e_g^2$ (4) $t_{2g}^0 e_g^4$

82. Which of the following statements are incorrect for $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

- (A) It is square planar complex
- (B) It is used in cancer treatment
- (C) It is mononuclear and heteroleptic complex
- (D) It is inner orbital complex
- (E) It shows attraction in magnetic field

Choose the correct answer from the options given below

- (1) A,B,C only
- (2) D, E only
- (3) A, B, E only
- (4) B, D, E only

83. Amongst the following, the total number of species which shows geometrical isomerism is $[\text{Co(en)}_2\text{Cl}_2]^+$, $[\text{Cr(NH}_3)_4\text{C}_2\text{Cl}]$, $[\text{Pt(NH}_3)_2\text{Cl}_2]$, $[\text{Co(NH}_3)_5(\text{NO}_2)]\text{Cl}$, $[\text{Pt(en)}_3]^{4+}$

- (1) 2 (2) 4
(3) 1 (4) 3

84. **Assertion :** Stability constant of the complexes $[\text{Cu(NH}_3)_4]^{2+}$ and $[\text{Cu(CN)}_4]^{2-}$ are 4.5×10^{11} and 2.0×10^{27} respectively.

Reason : Greater the strength of ligand, greater is the stability of the complex formed.

- (1) Both Assertion & Reason are true but Reason is NOT the correct explanation of Assertion
(2) Assertion is true but Reason is false
(3) Assertion is false but Reason is true
(4) Both Assertion and Reason are true & Reason is the correct explanation of Assertion

85. Match List-I with list-II

	List-I		List-II
(A)	$\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$	(I)	Bidentate ligand
(B)	SCN^-	(II)	Monodentate ligand
(C)	$\text{NH}_2 - \text{NH}_2$	(III)	π -acid ligand.
(D)	CO TG: @Chalnaayaaar	(IV)	Ambidentate ligand.

Choose the correct answer from options given below:

- (1) A-I, B-IV, C-II, D-III
(2) A-III, B-I, C-IV, D-II
(3) A-IV, B-I, C-II, D-III
(4) A-II, B-I, C-IV, D-III

SECTION-B (CHEMISTRY)

86. In a body centred cubic unit cell of a metal edge length is 4×10^{-10} m. The diameter of the atom will be :-

- (1) 3.46×10^{-10} m
(2) 2×10^{-10} m
(3) 1×10^{-10} m
(4) 1.732×10^{-10} m

87. The total number of octahedral void(s) per atom present in a cubic close packed structure is :-

- (1) 1 (2) 2 (3) 3 (4) 4

88. For antifluorite type solid $r_+ + r_-$ is equal to:

- (1) a (2) $\frac{a}{2}$
(3) $\frac{a\sqrt{3}}{4}$ (4) $\frac{a\sqrt{3}}{2}$

89. The number of unit cells in 58.5 g of NaCl is nearly:-

- (1) 6×10^{20} (2) 3×10^{22}
(3) 1.5×10^{23} (4) 0.5×10^{24}

90. Which of the following is incorrect for lead storage battery:

- (1) Electrolyte is 38% H_2SO_4 solution.
(2) During discharging conc. of H_2SO_4 decreases.
(3) This is a secondary cell.
(4) During charging PbSO_4 is oxidised into PbO_2 at cathode.

91. In the following reaction $\text{A} \rightarrow \text{B} + \text{C}$, rate constant is 0.001 Ms^{-1} . If we start with 1 M of A then concentration of A and B after 10 minutes are respectively

- (1) 0.5 M, 0.5 M (2) 0.6 M, 0.4 M
(3) 0.4 M, 0.6 M (4) None of these

92. In a certain gaseous reaction between X and Y, $X + 3Y \rightarrow XY_3$. The initial rates are reported as follows:

[X]	[Y]	Rate
0.1 M	0.1 M	0.002 Ms^{-1}
0.2 M	0.1 M	0.002 Ms^{-1}
0.3 M	0.2 M	0.008 Ms^{-1}

The rate law is :

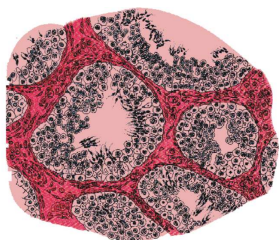
- (1) $r = K[X][Y]^3$ (2) $r = K[X]^0[Y]^2$
 (3) $r = K[X][Y]$ (4) $r = [X]^0[Y]^3$
93. For a zero order reaction $t_{75\%} = 30 \text{ min}$. Find $t_{50\%}$ -
 (1) 15 min (2) 20 min
 (3) 30 min (4) 75 min
94. For the reaction $2A \rightarrow B + 3C$;
 if $-\frac{d[A]}{dt} = k_1[A]^2$; $\frac{d[B]}{dt} = k_2[A]^2$; $\frac{d[C]}{dt} = k_3[A]^2$ the correct relation between k_1 , k_2 and k_3 is :-
 (1) $k_1 = k_2 = k_3$ (2) $2k_1 = k_2 = 3k_3$
 (3) $4k_1 = k_2 = 3k_3$ (4) $\frac{k_1}{2} = k_2 = \frac{k_3}{3}$
95. If AgNO_3 solution is added in excess to 1 M solution of $\text{CoCl}_3 \cdot x\text{NH}_3$ one mole of AgCl is formed. What is value of x ?
 (1) 1 (2) 2 (3) 3 (4) 4
96. Select the incorrect statement about metal carbonyl complex compounds :
 (1) Metal carbon bond in metal carbonyls possess both σ and π character
 (2) Due to synergic bonding metal carbon bond becomes weak
 (3) Due to synergic bonding carbon oxygen bond strength decreases
 (4) In metal carbonyls extent of synergic bonding will increase with increase in negative charge on central metal ion

97. The correct order of C – C bond length in the following compound is :-
 (I) C_2F_4 (II) C_2H_4 (III) $[\text{PtCl}_3(\text{C}_2\text{H}_4)]^{-1}$
 (1) $\text{I} > \text{II} > \text{III}$
 (2) $\text{I} < \text{II} < \text{III}$
 (3) $\text{I} > \text{III} > \text{II}$
 (4) $\text{I} > \text{II} = \text{III}$
98. The correct order for the wavelength of absorption in the visible region is
 (1) $[\text{Ni}(\text{NO}_2)_6]^{4-} < [\text{Ni}(\text{NH}_3)_6]^{2+} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
 (2) $[\text{Ni}(\text{NO}_2)_6]^{4-} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{NH}_3)_6]^{2+}$
 (3) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{NH}_3)_6]^{2+} < [\text{Ni}(\text{NO}_2)_6]^{4-}$
 (4) $[\text{Ni}(\text{NH}_3)_6]^{2+} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{NO}_2)_6]^{4-}$
99. **Assertion** : $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is strongly paramagnetic whereas $[\text{Fe}(\text{CN})_6]^{3-}$ is weakly paramagnetic.
Reason : $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ are outer orbital complexes involving sp^3d^2 hybridisation and are paramagnetic corresponding to five and one unpaired electrons.
 (1) Both Assertion & Reason are true but Reason is NOT the correct explanation of Assertion
 (2) Assertion is true but Reason is false
 (3) Assertion is false but Reason is true
 (4) Both Assertion and Reason are true & Reason is the correct explanation of Assertion
100. Which type of isomerism(s) is/are exhibited by complex $[\text{CrCl}_2(\text{H}_2\text{O})_2(\text{NH}_3)_2]\text{Br}$.
 (A) Ionisation isomerism
 (B) Hydrate isomerism
 (C) Linkage isomerism
 (D) Geometrical isomerism
 (E) Optical isomerism
 Choose the correct option from given option:
 (1) A, B, C only (2) A, D, E only
 (3) A, D only (4) D, F only

Topic : MOLECULAR BASIS OF INHERITANCE (TRANSCRIPTION TO HUMAN GENOME PROJECT), COMMUNITY INTERACTIONS TO ECOSYSTEM [BACK UNIT] : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH
TG: @Chalnaayaaar

SECTION-A (BIOLOGY-I)

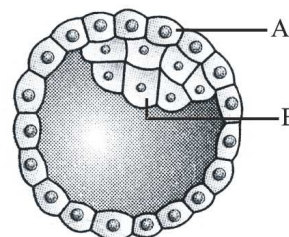
101. Which phase of oogenesis runs during menstrual cycle in human ?
- (1) Multiplication phase
 - (2) Yolk sunthesis
 - (3) Maturation phase
 - (4) Growth phase
102. Identify the structure shown below and its location in the human, and select the right option.



	Structure	Site of occurrence
(1)	Testis	Abdominal cavity
(2)	Penis	Abdominal cavity
(3)	Seminiferous tubules	Testis
(4)	Prostate	Below the urinary bladder

103. Which of following statement is incorrect ?
- (1) Each ovary is about to 2 to 4 cm in length and is connected to the pelvic wall and uterus by ligaments.
 - (2) The shape of the uterus is like an inverted pear.
 - (3) Oogenesis is initiated at puberty stage when a couple of million primary follicles are formed with in each ovary.
 - (4) First meiotic division of oogenesis is an unequal division resulting in formation of a large haploid secondary oocyte and a tiny first polar body.

104. The chances of fertilisation are very high during which days of menstruation cycle ?
- (1) First 5 days
 - (2) Last 7 days
 - (3) from 10th to 17th day
 - (4) from 21st to 25th day
105. In a female undergoing tubectomy, which one of the following event will not occur ?
- (1) Menstruation cycle
 - (2) Ovulation
 - (3) Fusion of sperm & ovum
 - (4) Formation of Graafian follicle
106. Identify A and B and their respective functions:



Blastocyst

	A	B	Function	
			A	B
(1)	Trophoblast	Inner cell mass	Attach to the Endometrium	Differentiated as embryo
(2)	Trophoblast	Inner cell mass	Differentiated as embryo	Attach to the Endometrium
(3)	Mesoderm	Inner cell mass	Differentiated as embryo	Attach to the Endometrium
(4)	Ectoderm	Mesoderm	Attach to the myometrium	Differentiated as embryo

107. Fertilisation has following processes :-

(a) Plasmogamy (b) Karyogamy
(c) Syngamy (d) Amphimixis

Arrange these in correct sequence :

- (1) a → b → c → d (2) b → a → d → c
- (3) c → a → b → d (4) c → a → d → b

108. How many structures in the list given below have $(22 + x)$ chromosome ?

Polar Body, Oogonia, Primary Oocyte, Ovum, Secondary oocyte

- (1) Four (2) Three (3) Two (4) One

109. Which statement is correct ?

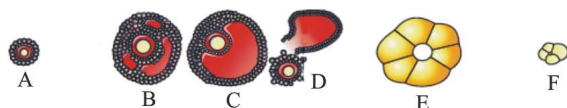
- (1) Glans penis is covered by a loose fold of skin called foreskin
 (2) The male accessory gland include paired seminal vesicle, paired prostate and paired bulbourethral gland
 (3) Both (1) and (2)
 (4) Breast are secondary sex organ

110. **Statement-1** :- The male sex accessory ducts include rete testis, vasa efferentia, epididymis and vas deferens.

Statement-2 :- Secretion of epididymis, vas-deferens, seminal vesicle and prostate are essential for maturation and motility of sperms. Read the above statements and choose correct option :-

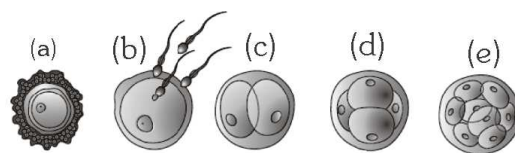
- (1) Statement-I and II both are correct.
 (2) Statement-I and II both are incorrect.
 (3) Only statement-I is correct.
 (4) Only statement-II is correct.

111. It is a diagrammatic presentation of developing follicle events, find out correct statement :-



- (1) A = Primary follicle, C = Secondary follicle, E = Corpus haemorrhagicum, F = Corpus luteum
 (2) B = Secondary follicle, D = Mature follicle, E = Corpus haemorrhagicum, F = Corpus luteum
 (3) A = Primary follicle, C = Mature follicle, E = Corpus luteum, F = Corpus albicans
 (4) A = Primary follicle, D = Mature follicle, E = Corpus luteum, F = Corpus albicans

112.



Which of the following is correct match for the above diagram?

- (1) (a) – Meiosis II completed
 (2) (b) – Cleavage initiated
 (3) (e) – Morula stage
 (4) (e) – Blastula stage

113. The funnel shaped part of fallopian tube is :-

- (1) Isthmus
 (2) Ampulla
 (3) Infundibulum
 (4) Intramural part

114. Which of the following layers of wall of the uterus exhibits strong contractions during delivery of baby ?

- (1) Outermost layer
 (2) Middle layer
 (3) Innermost layer
 (4) None of these

115. In breast of human female, an alveoli opens into:-

- (1) Mammary tubule
 (2) Mammary duct
 (3) Mammary ampulla
 (4) Lactiferous duct

116. According to the 2011 census report, the population growth rate was :-

- (1) Less than 1.7 percent
 (2) More than 1.7 percent
 (3) More than 2 percent
 (4) Less than 2 percent

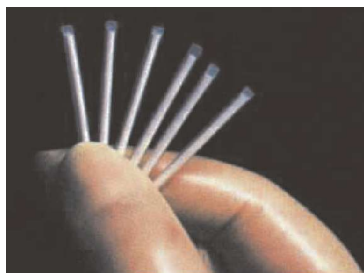
117. Which of the following is not an IUD :-

- (1) Lippes loop (2) LNG-20
(3) CuT (4) Vaults

118. In which of the following contraceptive method chances of failure are high ?

- (1) Withdrawal
(2) IUDs
(3) Oral contraceptives
(4) Condoms

119. Identify the contraceptive device shown below as well as the related right place of its implantation into a woman and select the correct option for the two together :



TG: @Chalnaayaaar

	Contraceptive device	Site of implant
(1)	LNG – 20	Fallopian tube
(2)	Lippes loop	Uterine wall
(3)	Implants	Subcutaneous
(4)	Multiload 375	Uterine wall

120. Match the column-A with column-B, and choose the correct option :-

Column-A	Column-B
(A) Barrier method	(i) Tubectomy
(B) Oral pills	(ii) Condoms
(C) IUD's	(iii) Prevents ovulation
(D) Sterilisation	(iv) Cu-T

- (1) A-iv, B-ii, C-iii, D-i
(2) A-ii, B-iii, C-i, D-iv
(3) A-ii, B-iii, C-iv, D-i
(4) A-i, B-ii, C-iv, D-iii

121. The prickly pear cactus introduced in to Australia in the early 1920s caused havoc by spreading rapidly in to millions of hectares of rangeland Because :-

- (1) There was no competition
(2) Due to less interspecific competition
(3) Absence of Natural predator
(4) Availability of high nutrient

122. Which of the following is a conduit for energy transfer across trophic levels ?

- (1) Mutualism
(2) Parasitism
(3) Predation
(4) Proto cooperation

123. **Assertion** : Cuscuta is a parasitic plant that is commonly found growing on hedge plants.

Reason : It lost its chlorophyll, leaves and derive its nutrition from host plant.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

124. Which adaptation of parasite do not help to sustain their life in host body ?

- (1) Loss of unnecessary sense organ
(2) Presence of adhesive organ or suckers to cling the host
(3) Loss of digestive system
(4) Low reproductive capacity

125. Annually how much amount of carbon is fixed by photosynthesis in biosphere?

- (1) 4×10^{23} kg
- (2) 4×10^{13} kg
- (3) 170 billion tons
- (4) 71 kg

126. Which of the following is not a functional unit of an ecosystem ?

- (1) Nutrient cycling
- (2) Energy flow
- (3) Productivity
- (4) Stratification

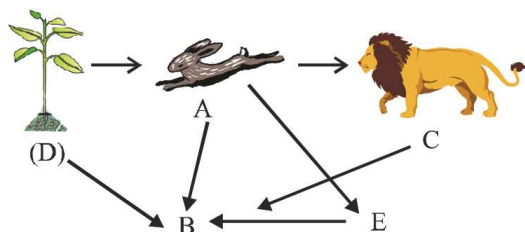
127. On the basis of ecology which is not essential for self sustaining ecosystem :

- (1) Producer
- (2) Consumer
- (3) Decomposer
- (4) All of above

128. Ecologically most relevant environmental factor –

- (1) Temperature
- (2) Water
- (3) Soil
- (4) Light

129. In the below diagram 'B' is :



- (1) Primary consumer
- (2) Decomposer
- (3) Top consumer
- (4) Producer

130. Which of the following can be considered as herbivores of aquatic ecosystem :-

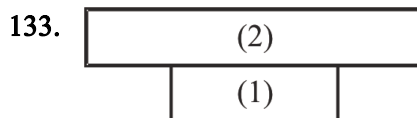
- (1) Earthworm
- (2) Molluscs
- (3) Fungus
- (4) Vallisneria

131. The number of primary producers within a specified area would be maximum in :-

- (1) Grassland ecosystem
- (2) Forest ecosystem
- (3) Pond ecosystem
- (4) Deserts

132. Which one of the following animals may occupy more than one trophic levels in the same ecosystem at the same time ?

- | | |
|-----------|-------------|
| (1) Snake | (2) Sparrow |
| (3) Wolf | (4) Deer |



Above pyramid represent.

- (1) Pyramid of grassland on the basis of biomass.
- (2) Pyramid of grassland on the basis of number.
- (3) Pyramid of aquatic ecosystem on basis of biomass
- (4) Pyramid of energy

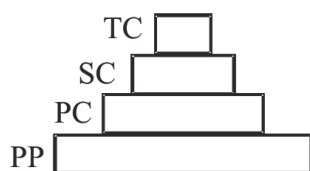
134. In the equation

$$\boxed{GPP - R = NPP}$$

R here is –

- (1) Respiratory quotient
- (2) Respiratory loss
- (3) Reproductive capacity
- (4) Photosynthetically active radiation

135.



Sunlight 10,000 J

What would be amount of energy trapped at top trophic level ?

- (1) 10 J
- (2) 1 J
- (3) 100 J
- (4) 1000 J

SECTION-B (BIOLOGY-I)

136. Which of the following statement is incorrect ?

- (1) The male germ cells undergo meiotic divisions finally leading to sperm formation
- (2) The functions of male sex accessory ducts and glands are maintained by the testicular hormones
- (3) In fact, the presence or absence of hymen is not a reliable indicator of virginity
- (4) Sperm and ovum formation ceases around the age of fifty years.

137. The edges of the infundibulum posses finger like projections called fimbriae which helps in:

- (1) Collection of sperms
- (2) Collection of ovum after ovulation
- (3) To facilitate implantation
- (4) To pass nutrition towards ovum

138. The first sign of growing foetus may be noticed by :-

- (1) Movement of foetus
- (2) Appearance of hair on head
- (3) Listening to the heart sound carefully through the stethoscope
- (4) Formation of limbs

139. Match the Column-A with Column-B :-

	Column-A		Column-B
(A)	Mons pubis	(i)	Fleshy folds of tissue
(B)	Labia majora	(ii)	Paired folds of tissue
(C)	Labia minora	(iii)	Finger-like structure
(D)	Clitoris	(iv)	Cushion of fatty tissue

- (1) A-ii, B-iv, C-i, D-iii
- (2) A-iv, B-ii, C-i, D-iii
- (3) A-iv, B-i, C-ii, D-iii
- (4) A-iii, B-iv, C-ii, D-i

140. A temporary endocrine gland.....formed after ovulation in ovary and secretes.....hormone :-

- (1) Corpus albicans & estrogen respectively
- (2) Corpus luteum & LH respectively
- (3) Corpus albicans & progesterone respectively
- (4) Corpus luteum & progesterone respectively

141. How many of the following structures are present only in male ?

Vas deferens, Seminiferous tubule, Mammary gland, Anus, Urethra, Epididymis, Bartholin gland, Clitoris.

- (1) Five
- (2) Three
- (3) Seven
- (4) Four

142. Egg is liberated from ovary and enters the fallopian tube in :-

- (1) Secondary oocyte stage
- (2) Primary oocyte stage
- (3) Oogonial stage
- (4) Mature ovum stage

143. How much percent of total sperms ejaculated by a human male during a coitus must show vigorous motility, for normal fertility ?

- (1) 60% atleast
- (2) 40% atleast
- (3) 24% atleast
- (4) 50% atleast

144. Which of the following are natural contraceptive methods :-

- (A) Periodic abstinence
- (B) Coitus interruptus
- (C) Lactational amenorrhoea
- (D) M.T.P.

- (1) A, B and D
- (2) A, C, D
- (3) A, B, C
- (4) B, D

145. Which of the following set involves only invitro fertilization methods ?

- (1) ZIFT, IUT (2) GIFT, IUT, ICSI
- (3) GIFT, ZIFT, IUT (4) GIFT, ICSI

146. Major conduit of energy flow in :-

	Terrestrial ecosystem	Aquatic ecosystem
(1)	GFC	DFC
(2)	DFC	GFC
(3)	DFC	DFC
(4)	GFC	GFC

147. Which is the potent force in organic evolution :

- (1) Inter-species competition
- (2) Predation
- (3) Intra-species competition
- (4) Mutualism

148. **Assertion** : Five closely related species of warblers living on the same tree can avoid competition and co-exist.

Reason : They exhibit behavioral differences in their foraging activities.

- (1) If both Assertion and Reason are true and the reason is the correct explanation of the assertion
- (2) If both Assertion and Reason are true but the reason is not the correct explanation of the assertion
- (3) If Assertion is true statement but Reason is false
- (4) If both Assertion and Reason are false statements

149. Primary producers in aquatic ecosystem are :-

- (a) Phytoplankton (b) Algae
- (c) Higher plants (d) Fungi

Choose the correct option :-

- (1) a + b + d
- (2) a + b only
- (3) Only a
- (4) a + b + c

150. Amount of living material and nutrients present in different trophic levels and soils at any given time are called respectively :-

- (1) Standing state and standing crop
- (2) Standing crop and standing state
- (3) Standing state and standing quality
- (4) Biomass and standing crop

Topic : MOLECULAR BASIS OF INHERITANCE (TRANSCRIPTION TO HUMAN GENOME PROJECT), COMMUNITY INTERACTIONS TO ECOSYSTEM [BACK UNIT] : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH

TG: @Chalnaayaaar

SECTION-A (BIOLOGY-II)

151. Which of the following ecosystem has the highest annual net primary productivity ?

- (1) Tropical deciduous forest
- (2) Temperate deciduous forest
- (3) Tropical rain forest
- (4) Temperate coniferous forest

152. Which one is correct :-

	I		II
(1)	Formation of dark coloured amorphous substance	(i)	Fragmentation
(2)	Water soluble inorganic nutrients go down into the soil horizon	(ii)	Humification
(3)	release of inorganic nutrient by microbial activities	(iii)	Mineralisation
(4)	Break down detritus into smaller particles	(iv)	Leaching

153. Rate of breakdown of complex organic matter will be fast in which of the following ?

- (1) Hot and humid climate
- (2) Anaerobic conditions
- (3) Area of high temperature and less humidity
- (4) Area of low temperature and high humidity

154.

In above diagram B is :-

- (1) Primary consumer
- (2) Secondary consumer
- (3) Top consumer
- (4) Producer

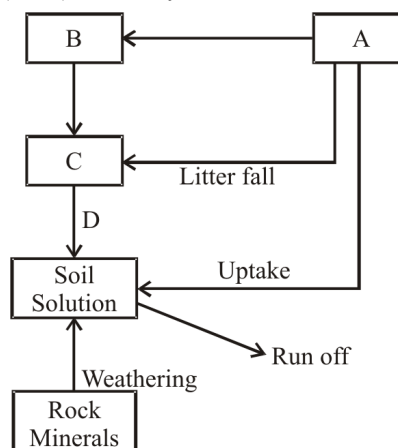
155. Animals use light as cues for –

- (1) Their foraging activities
- (2) Their reproductive activities
- (3) Their migratory activities
- (4) All of the above

156. Connell's Elegant field experiment done on –

- (1)
- (2)
- (3)
- (4)

157. Given below is a simplified model of phosphorus cycling in a terrestrial ecosystem with four blanks (A-D). Identify the blanks :-

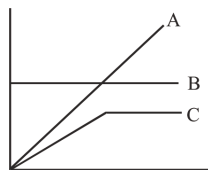


	A	B	C	D
(1)	Producers	Consumers	Detritus	Decomposition
(2)	Producers	Detritus	Consumers	Decomposition
(3)	Consumers	Producers	Detritus	Decomposition
(4)	Weathering	Detritus	Producers	Consumers

158. Plants capture only _____ percent of PAR which sustains the whole biosphere.

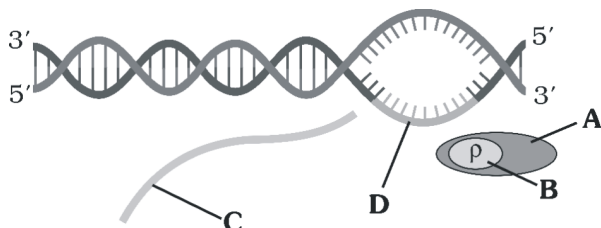
- (1) 1–5% (2) 2–10% (3) <50% (4) >50%

159. Identify A, B & C in the given figure :-



	A	B	C
(1)	Conformes	Regulator	Partial Regulator
(2)	Partial Regulator	Conformes	Regulator
(3)	Regulator	Partial Regulator	Conformes
(4)	Conformes	Partial Regulator	Regulator

160. Carefully observe the diagram below and identify the items A, B, C and D.



- (1) A-DNA polymerase, B-rho factor, C-RNA, D-Template strand
 (2) A-sigma factor, B-RNA polymerase, C-RNA, D-Template strand
 (3) A-RNA polymerase, B-rho factor, C-RNA, D Template strand
 (4) A-Primase, B-rho factor, C-RNA, D-coding strand

161. How many statements are correct about polycistronic mRNA?

- (A) Codes for more than one polypeptide.
 (B) Commonly found in eukaryotes
 (C) More than one stop codons present.
 (D) More than one ribosomes may be attached to such mRNA at same time

- (1) Two (2) One (3) Four (4) Three

162. Identify the enzyme A, B & C in the given diagram :-



	A	B	C
(1)	DNA polymerase	RNA polymerase	RNA polymerase
(2)	RNA polymerase	RNA polymerase	Peptidyl transferase
(3)	DNA polymerase	RNA polymerase	Ribozyme
(4)	RNA polymerase	DNA polymerase	Peptidyl transferase

163. Given below is sequence of the processed mRNA ready for translation 5'-AUG CUA UAC UAA CUG CCA UGC UAG-3'. How many amino acids will be presented in polypeptide chain corresponding to this mRNA :-

- (1) 7 (2) 8 (3) 6 (4) 3

164. What would happen if in a gene encoding a polypeptide of 50 amino acids, 30th codon (UAU) is mutated to UAA :-

- (1) A polypeptide of 29 amino acids will be formed
 (2) Two polypeptides of 24 and 25 amino acids will be formed
 (3) A polypeptide of 49 amino acids will be formed
 (4) A polypeptide of 30 amino acids will be formed

165. During protein synthesis formation of peptide bonds takes place between :-

- (1) $-\text{NH}_2$ group of 'P' site amino acid and COOH group of 'A' site amino acid
 (2) $-\text{COOH}$ group of 'P' site amino acid and NH_2 group of 'A' site amino acid
 (3) $-\text{NH}_2$ group of 'P' site amino acid and $-\text{NH}_2$ group of 'A' site amino acid.
 (4) $-\text{CH}_3$ group of 'P' site amino acid and $-\text{COOH}$ group of A site amino acid

174. Identify the correct sequence in DNA fingerprinting ?

- (a) Gel electrophoresis
- (b) Hybridization using VNTR probe
- (c) Autoradiography
- (d) Southern blotting

(1) a, b, c, d (2) b, d, c, a

(3) a, d, b, c (4) b, a, c, d

175. Human Genome Project was closely associated with the rapid development of a new area in biology called :-

- (1) Genomics
- (2) Genetic Engineering
- (3) Bioinformatics
- (4) Biotechnology

176. In HGP, the blind approach of simply sequencing the whole set of genome that contained all the coding and non-coding sequences, is known as

- (1) Expressed Sequence Tags
- (2) ELSI
- (3) Single nucleotide polymorphism
- (4) Sequence Annotation

177. **Assertion** : The codon UUU always codes phenyl alanine amino acid.

Reason : Codon is unambiguous in nature

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

178. Operon model for the regulation of gene action in *E.coli* was proposed by :-

- (1) Francis Jacob & Jacques Monod
- (2) Beadle & Tatum
- (3) Temin & Baltimore
- (4) Watson & Crick

179. In RNA splicing non-coding parts are removed with help of ___A___ and coding parts joined together with help of ___B___. Identify A & B in given statement.

	A	B
(1)	RNA ligase	DNA ligase
(2)	Spliceosome complex	RNA ligase
(3)	Sigma factor	DNA ligase
(4)	RNA polymerase	RNA ligase

180. Which of the following statement is incorrect :-

- (1) Chromosome-1 has maximum genes.
- (2) Chromosome-x has minimum genes.
- (3) Largest gene is of dystrophin (2.4 million base)
- (4) Sequence of chromosome-1 was completed in may-2006

181. One Codon specify only one amino acid and not other, this property of genetic code is known as :

- (1) Degeneracy of genetic code
- (2) Commaless
- (3) Non-overlapping
- (4) Unambiguous

182. In lac-operon, lactose is transported into the cells through the action of which enzyme.

- (1) β - galactosidase
- (2) Transacetylase
- (3) Permease
- (4) Peptidyl transferase

192. Which of the following RNA is synthesized by RNA polymerase III in eukaryotes:-

- (1) 5.8s rRNA and hnRNA
- (2) snRNA and 5s rRNA
- (3) tRNA and 28s rRNA
- (4) tRNA and hnRNA

193. **Assertion** : The split gene arrangements represent probably an ancient feature of the genome.

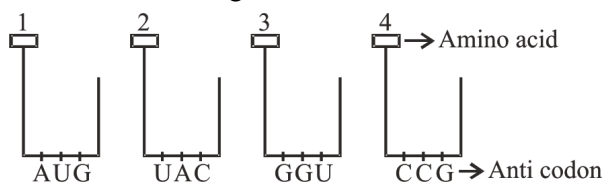
Reason : The presence of introns is reminiscent of antiquity and the process of splicing represent the dominance of RNA world.

- (1) If both assertion & reason are true & the reason is a **correct** explanation of the assertion.
- (2) If both assertion & reason are true but reason is **not** a correct explanation of the assertion.
- (3) If assertion is **true** but the reason is **false**
- (4) If both assertion & reason are **false**

194. A eukaryotic gene was introduced into bacteria along with suitable promoter but it was found that this gene did not form similar protein in bacteria. It is due to the :-

- (1) Absence of RNA polymerase.
- (2) Absence of origin of replication.
- (3) Absence of splicing mechanism.
- (4) Absence of 80s ribosomes in bacteria.

195. Find the sequence of binding of the following aminoacyl t-RNA complexes during translation to a mRNA transcribed by a DNA segment having base sequence 3' TACATGGGTCCG 5'. Choose the answer showing the correct order of t-RNA :-



- (1) 2, 1, 3, 4
- (2) 1, 2, 3, 4
- (3) 1, 2, 4, 3
- (4) 2, 1, 4, 3

196. Change in chromosome number is known as :-

- (1) Chromosomal aberration
- (2) Genomatic mutation
- (3) Point mutation
- (4) Gene mutation

197. Haploids are more suitable for mutation studies than the diploids because :-

- (1) Haploids are more abundant in nature than diploid
- (2) All mutations whether dominant or recessive are expressed in haploids
- (3) Haploids are reproductively more stable than diploids
- (4) Mutagen penetrate in haploids more effectively than in diploids

198. Which of the following statements concerning the regulatory genes (i), associated with the lac operon are incorrect ?

I. mRNA is transcribed from the i gene whether lactose is present or not.

II. mRNA is transcribed from the i gene only when the lactose is present.

III. mRNA is transcribed from the i gene only when the lactose is not present.

IV. Lactose inhibits the translation of i gene mRNA.

- (1) I and II
- (2) II and III
- (3) III and IV
- (4) II, III and IV

199. Match the column I with II and choose the correct answer :

Column-I		Column-II	
A	Frame shift deletion	i	Cri-du-chat syndrome
B	Transversion-substitution	ii	Chronic myelogenous leukemia
C	Terminal deletion	iii	Thalassemia
D	Reciprocal translocation	iv	Sickle cell Anemia

- (1) A - ii, B - iii, C - iv, D - i
 (2) A - iii, B - iv, C - i, D - ii
 (3) A - ii, B - iv, C - i, D - iii
 (4) A - iii, B - i, C - iv, D - ii

200. Which of the following represent correct set of stop codons ?

- (1) UUC, UUA, UUG
 (2) UCU, UCA, UCG
 (3) UAA, UAG, UGA
 (4) UGG, UGC, UGU



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